Software Team

College0
Software Requirements Specification
For College Management System

Version 1.0

Revision History

Date	Version	Description	Author
24/Mar/2026	1.0	Wrote Introduction, Purpose, Scope sections and overall formatting	Gaurab Pun
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Software Requirements Specification

1. Introduction

This Software Requirements Specification (SRS) document provides a detailed description of the College0 system, an AI-enabled online college program management system. The document outlines the system's features, user roles, and functional requirements. It serves as a guide for developers and stakeholders to understand how the system should behave and what functionalities it must include.

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements of the College0 system. It describes how the system should operate, including user interactions, system behavior, and constraints. This document is intended for developers, project team members, and instructors to ensure a clear understanding of the system requirements before implementation.

1.2 Scope

College0 is an AI-enabled college program management system designed to support multiple types of users, including registrars, instructors, students, and visitors. The system allows registrars to manage courses, instructors, and student records. Students can register for courses, view their academic progress, and provide feedback through reviews. Instructors can manage assigned courses, view student information, and assign grades. Visitors can browse general information and apply to become students or instructors.

Additionally, the system includes an AI-powered assistant that helps users by answering questions related to courses, requirements, and academic information. The system ensures structured academic processes such as course registration, grading, and performance monitoring.

1.3 Definitions, Acronyms, and Abbreviations

- **AI (Artificial Intelligence):** Technology used for the system's Q&A assistant and text moderation
- **GPA (Grade Point Average):** The numerical representation of a student's academic achievement.
- **LLM (Large Language Model):** An external AI service used to answer general queries when local information is unavailable.
- **RBAC (Role-Based Access Control):** A security mechanism that restricts system access based on the user's role (Registrar, Instructor, Student, or Visitor).
- **Taboo Words:** A registrar-managed list of prohibited terms used to filter student course reviews.
- **UC (Use Case):** A description of a system's behavior as it responds to a request from a user.
- **Vector DB:** A local database storing specific institutional information for the AI assistant to provide grounded, hallucination-free answers.

1.4 References

- **College0 Project Specification:** "An AI-enabled online college program management system" design document.
- **Rational Unified Process (RUP):** Software Requirements Specification Template.

1.5 Overview

- **Section 1 (Introduction)**: Defines the purpose, scope, and terminology of the College0 system.
- **Section 2 (Overall Description):** Provides a high-level overview of product perspective, user characteristics, and general constraints.

- **Section 3 (Specific Requirements):** Detailed functional requirements via 19 Use Cases and non-functional requirements such as security and performance.
- **Section 4 (Supporting Information):** Contains the index and architectural appendices.

2. Overall Description

College0, an AI-enabled online college program management system that is designed to support academic operations such as course registration, grading, class reviews, complaints, and academic records. It manages all the interactions between registrar, instructors, students, and visitors. The system will also include an AI-assistant where you can ask about classes and requirements. There are 4 user groups as mentioned before and they all have different accesses such as the registrars having full access who are able to do anything while students will only have access to classes , their grades, and class reviews. Some constraints of this program may include privacy of student academic records, strict academic policies such as GPA thresholds, and scalability. The program requirements can be grouped into functional, non-functional, and AI-specific requirements. The functional group may include admissions, class registration, grades, reviews/complaints. Non-functional group will be performance, usability, scalability and the AI-specific could be a text area for query handling.

2.1 Use-Case Model Survey

Actors	College0 System
Visitor	Apply as Student Apply as Instructor View public info
Student	Register for courses View academic records Submit reviews/complaints Ask AI questions
Instructor	Manage courses Submit grades Manage waitlist Ask AI questions
Registrar	Approve/Reject applications Set up semester Manage courses Handle complaints

Use Cases:

- Admission & Account Creation
  - Visitor can apply to be a student or teacher
  - First login and changing password
- Semester Management
  - 4 periods managed by the registrars which include, Class set-up period, course registration period, class running period, grading period
  - Registrar set-up the classes, time, instructors and size
- Course Registration
  - Students enroll in courses depending on the restrictions
  - Special registration period
- Class-Running Period

    - Students won't be able to register for courses during the class running period
    - Warnings will be given to students and instructors
- Grading Period
    - Instructors submit grades and obtains a class GPA
    - Evaluate academic standings of the class and warnings/terminations/honors will be given
    - Apply for graduation
- Review/Complaints
    - Students can write reviews for the course
    - Students and/or Instructors can file complaints to the registrar
    - Issue warnings/suspensions for students/instructors depending on their situation
- AI Feature
    - All users can ask questions
    - Answers should be found in local information and if not, send to LLM with warning

2.2 Assumptions and Dependencies

- Standard academic structure, 4.0 GPA scale, semesters
- AI feature relies on a vector database to store local information

3. Specific Requirements

Visitor:

- Browse the public dashboard (program intro, highest/lowest rated classes, top GPA students).
- Apply to become a student.
- Apply to become an instructor.
- Ask general AI questions about classes and program requirements.

Student:

- Log in and change password on first login.
- Complete onboarding tutorial of the system(new students only).
- View personal dashboard (courses, grades, GPA, warnings, honor roll status).
- Register for 2-4 courses during registration period (no time conflicts, class not full).
- Join a course waitlist if a class is full.
- Retake a course previously failed (F grade).
- Write reviews and assign star ratings (1–5) to enrolled courses (before grades are posted).
- Ask AI questions about their own enrolled courses.
- File complaints against other students or instructors.
- Use honor roll status to remove a warning.
- Apply for graduation after completing 8 classes.
- Receive warnings, suspensions, or termination based on GPA and conduct rules.

Instructor:

- Log in and view a personal dashboard with current assigned courses.
- View basic info and academic records of students in their courses.
- Admit waitlisted students into their course.
- Assign grades to all enrolled students during the grading period.
- File complaints against students (requesting warning or de-registration).
- Ask AI questions about students in their current courses.
- Receive warnings for low course ratings, incomplete grading, abnormal class GPA, or cancelled courses.
- Be suspended if all courses are cancelled or 3 warnings are accumulated.

Registrar:

- Do everything any other user type can do.
- Accept or reject student and instructor applications (with justification if deviating from GPA/quota rules).
- Set up classes, schedules, instructors, and class sizes each semester.
- Manage semester period transitions (setup → registration → running → grading).
- Manage the taboo word list for review moderation.
- Admit or process waitlist decisions.
- View all course reviews including reviewer identities.
- Review and question instructors whose class GPA falls outside the 2.5–3.5 range.
- Process all complaints from students and instructors.
- Approve or reject graduation applications.
- Issue warnings, suspensions, fines, or terminations to students and instructors.
- Access and modify any record in the system.

3.1 Use-Case Reports (UC - “Use Case)

UC-01: View Public Dashboard

Actors	Visitor, Student, Instructor, Registrar
Trigger	User navigates to the home page
Precondition	None. No authentication is required.
Main Flow	The system renders the public dashboard, displaying a general introduction to the program, a list of the highest-rated classes ranked by average star rating, a list of the lowest-rated classes, and a leaderboard of students with the highest cumulative GPA. All four sections are visible without logging in.
Alternative Flow	None
Post Conditions	The user has successfully viewed all public-facing program information.
Functional Constraints	All user types, including unauthenticated visitors, have full access to this page. Class ratings are computed as the mean of all submitted student star ratings. Student identities on the GPA leaderboard are visible to all users.

UC-02: Apply to be Student

Actors	Visitor, Registrar
Trigger	A Visitor selects the option to apply as a student.
Precondition	The user is a Visitor with no existing account in the system.
Main Flow	The Visitor completes and submits an application form with their personal information and academic background. The system records the application and notifies the Registrar. The Registrar reviews the application and makes an accept or reject decision. If accepted, the system automatically generates a unique student ID and a temporary password, then notifies the applicant with those credentials. If rejected, the system notifies the applicant of the decision.
Alternative Flow	If the Registrar accepts an applicant whose GPA is at or below 3.0, or whose acceptance would exceed the program quota, the system blocks the decision and requires the Registrar to enter a written justification before it can be saved. The same justification requirement applies if the Registrar rejects an applicant whose GPA exceeds 3.0 and quota is still available.
Post Conditions	The Visitor is either enrolled as a Student with login credentials, or has been formally notified of their rejection.
Functional Constraints	Any applicant with a GPA above 3.0 and an available program quota slot must be accepted. Any applicant who does not meet that threshold must be rejected unless the Registrar provides written justification for the deviation. Accepted students receive a unique student ID and a temporary password that must be changed on their first login.

UC-03: Apply to be Instructor

Actors	Visitor, Registrar
Trigger	A Visitor selects the option to apply as an instructor.
Precondition	The user is a Visitor with no existing instructor account in the system.

Main Flow	The Visitor submits an instructor application including their professional credentials and teaching background. The system records the application and notifies the Registrar. The Registrar reviews the application and issues an approval or rejection decision. If approved, the system creates an instructor account and the Registrar assigns the new instructor to one or more classes. If rejected, the applicant is notified without any required explanation.
Alternative Flow	None.
Post Conditions	The Visitor is either an active Instructor with an account and assigned classes, or has been notified of their rejection.
Functional Constraints	The Registrar is not required to provide justification for instructor application decisions in either direction.

UC-04: Login

Actors	Student, Instructor, Registrar
Trigger	A registered user enters their credentials on the login screen.
Precondition	The user has a valid account and credentials in the system.
Main Flow	The user enters their username and password. The system authenticates the credentials and routes the user to the home page appropriate to their role, where they can access all features available to them.
Alternative Flow	If it is a student's first login, the system detects the temporary password status and forces the student to set a new password before any other action. After the password is changed, the system presents a mandatory onboarding tutorial before granting access to the dashboard.
Post Conditions	The user is authenticated and has been directed to their role-specific home page.
Functional Constraints	First-login password change is mandatory for all new students and cannot be skipped. Each user role is routed to a different home page with access scoped to their permissions.

UC-05: Class Setup

Actors	Registrar
Trigger	The Registrar opens the Class Setup period to begin configuring courses.
Precondition	The system is in the Class Setup period of a new semester, initiated by the Registrar.
Main Flow	The Registrar creates one or more classes by specifying the class name, scheduled meeting time, maximum enrollment size, and the instructor assigned to teach it. The system saves each class configuration and makes them available for review. Once all classes are configured, the Registrar closes the setup period and the system transitions to the Course Registration period.
Alternative Flow	None.
Post Conditions	All classes for the semester are defined and ready for student registration.
Functional Constraints	Every class must have a designated instructor, a time slot, and a maximum class size set before it can be published to students. Only the Registrar has the authority to

	create classes and assign instructors.

UC-06: Register for Courses

Actors	Student
Trigger	A Student browses and selects courses during the Course Registration period.
Precondition	The system is in the Course Registration period and the Student is actively matriculated with no holds on their account.
Main Flow	The Student browses the list of available classes and selects between two and four courses they wish to take. The system validates that none of the selected classes have overlapping time slots, and that none of the selected classes have reached their maximum enrollment. If both checks pass, the system confirms the registration and enrolls the student.
Alternative Flow	If a selected class is at full capacity, the system informs the student and places them on that class's waitlist rather than enrolling them directly. If the student previously received an F in a course, the system permits them to re-register for that same course as a retake.
Post Conditions	The student is formally enrolled in two to four courses, or is placed on waitlists for any full courses they selected.
Functional Constraints	Students must register for a minimum of two and a maximum of four courses per semester. Time conflicts between selected courses are not permitted under any circumstance. A student may only retake a course they previously failed with an F grade. Admitting a student from the waitlist is exclusively at the course instructor's discretion.

UC-07: Manage Waitlist

Actors	Instructor
Trigger	An Instructor reviews the waitlist for one of their courses.
Precondition	The system is in the Course Registration period and at least one student is on the waitlist for the Instructor's course.
Main Flow	The Instructor navigates to the waitlist for their course and reviews the students waiting for a spot. The Instructor selects one or more students to admit. The system checks that the class size limit has not been reached and enrolls the selected students.
Alternative Flow	If the class is already at its maximum size, the system prevents further admissions regardless of the instructor's selection.
Post Conditions	The selected waitlisted students have been enrolled in the course, or the waitlist remains unchanged if capacity is full.
Functional Constraints	Only the course's assigned instructor has the authority to admit students from the waitlist. No other actor, including the Registrar, can bypass this by admitting students on the instructor's behalf through this use case.

UC-08: Course Cancellation and Special Registration

Actors	System (automated), Student
Trigger	The system transitions into the Class Running period.
Precondition	The Course Registration period has closed and the system is transitioning into the Class Running period.
Main Flow	At the start of the Class Running period, the system automatically checks the enrollment count of every course. Any course with fewer than three enrolled students is cancelled. The system notifies the instructor of each cancelled course and issues them a warning. Students enrolled in cancelled courses are notified and given a Special Registration period during which they may choose replacement courses from those still active. The system also checks that every enrolled student is registered in at least two courses, and issues a warning to any student who falls below that minimum.
Alternative Flow	If all of an instructor's courses for the semester are cancelled, the system flags that instructor as suspended, preventing them from teaching in the following semester.
Post Conditions	All under-enrolled courses have been cancelled, affected students have had the opportunity to re-register, and all applicable warnings have been issued.
Functional Constraints	The minimum viable class size is three students. Instructors receive one warning for each course that is cancelled. An instructor whose entire course load is cancelled is automatically suspended for the next semester. Students enrolled in fewer than two courses after cancellations receive a warning. All standard course registration rules from UC-07 continue to apply during the Special Registration period.

UC-09: Submit Course Review and Rating

Actors	Student
Trigger	An enrolled Student submits a written review and star rating for one of their courses.
Precondition	The Student is currently enrolled in the course and the instructor has not yet posted final grades for it.
Main Flow	The Student writes a review and selects a star rating between one and five. The system scans the review text against the active taboo word list maintained by the Registrar. If no taboo words are found, the review is published as written. The system then recalculates and updates the course's average star rating.
Alternative Flow	If the review contains one or two taboo words, those specific words are replaced with asterisks, the filtered review is published, and the student receives one warning. If the review contains three or more taboo words, the review is suppressed entirely and not shown to anyone, and the student receives two warnings.
Post Conditions	The review has been published (possibly filtered) or suppressed, the course average rating has been updated, and any applicable warnings have been applied to the student's record.
Functional Constraints	Only students currently enrolled in a course may submit a review for it. Reviews cannot be submitted after the instructor has posted final grades. Star ratings are integers from one to five. The identity of the reviewer is visible only to Registrars. If a course's average rating drops below 2.0, the system automatically issues a warning to that course's instructor.

UC-10: Manage Taboo Word List

Actors	Registrar
Trigger	The Registrar navigates to the content moderation settings.
Precondition	The Registrar is authenticated and has navigated to the system configuration panel.
Main Flow	The Registrar views the current list of taboo words configured in the system. The Registrar may add new words, edit existing entries, or remove words from the list. The system saves the updated list, which becomes immediately active for all subsequent review submissions.
Alternative Flow	None.
Post Conditions	The taboo word list has been updated and is active for future review moderation.
Functional Constraints	Only the Registrar has the authority to manage the taboo word list. The version of the taboo word list active at the time a review was submitted is the version used to evaluate that review, not any later updated version.

UC-11: Assign Grades

Actors	Instructor
Trigger	The system enters the Grading period and the Instructor opens their course roster to assign grades.
Precondition	The system is in the Grading period and the Instructor has at least one active course with enrolled students.
Main Flow	The Instructor navigates to their course roster and assigns a final grade to each enrolled student. Once all grades are submitted, the system records them and recalculates each student's semester GPA and cumulative GPA. The system then automatically triggers the post-grading review checks described in UC-13 and UC-14.
Alternative Flow	If the Grading period closes before the Instructor has assigned grades to all enrolled students, the system automatically issues the Instructor one warning for incomplete grading.
Post Conditions	All grades are recorded, GPA calculations are updated across the system, and post-grading checks have been triggered.
Functional Constraints	Every enrolled student in a course must receive a final grade before the Grading period closes. Instructors who fail to grade all students by the deadline automatically receive one warning.

UC-12: Post Grading Instructor Review

Actors	Registrar, Instructor
Trigger	The Grading period has ended and all course GPAs have been fully computed by the system.
Precondition	The Grading period closes and the system calculates final class GPAs for all courses.
Main Flow	The system evaluates the class GPA for every course. Any course whose class GPA falls above 3.5 or below 2.5 is flagged and the Registrar is alerted. The Registrar contacts the instructor of the flagged course to request a written justification for the grade distribution. If the Registrar finds the justification adequate, no further action is

	taken. If the justification is deemed inadequate, the Registrar may issue the instructor a warning or terminate their employment immediately.
Alternative Flow	None.
Post Conditions	All flagged instructors have been reviewed and appropriate disciplinary action has been applied or cleared.
Functional Constraints	Any class GPA outside the range of 2.5 to 3.5 must be reviewed by the Registrar. An inadequate or absent justification may result in a warning or immediate termination at the Registrar's discretion. An instructor who accumulates three total warnings is automatically suspended from teaching.

UC-13: Post Grading Student Status Update

Actors	System (automated)
Trigger	Semester grades have been submitted by all instructors and all GPA values have been fully computed and stored.
Precondition	The Grading period closes and all GPA calculations are finalized.
Main Flow	The system automatically evaluates every student's cumulative GPA and course history. Students meeting termination criteria are removed from the system. Students meeting warning criteria are notified and flagged accordingly. Students meeting honor roll criteria are labeled as honor roll students, and any existing warning on their record may be removed using that honor roll designation if the student chooses.
Alternative Flow	None.
Post Conditions	All student statuses have been updated, the honor roll has been published, and all terminations and warnings are reflected in student records.
Functional Constraints	A cumulative GPA below 2.0 results in automatic termination. Earning an F grade in the same course twice results in automatic termination. A cumulative GPA between 2.0 and 2.25 results in one warning and a mandatory registrar interview. A semester GPA above 3.75 earns a semester honor roll designation. A cumulative GPA above 3.5 after more than one completed semester earns an overall honor roll designation. Each honor roll designation may be redeemed to remove one existing warning.

UC-14: Apply for Graduation

Actors	Student, Registrar
Trigger	A Student submits a graduation application through the system
Precondition	The Student has completed at least eight classes in the program and is in good standing to submit an application.
Main Flow	The Student submits a formal graduation application. The Registrar reviews the student's full transcript to verify that all program-required courses have been completed. If all requirements are satisfied, the student is graduated, awarded a Bachelor's degree, and their account is removed from the active system.
Alternative Flow	If the Registrar finds that not all required courses have been completed, the graduation application is denied and the student receives one warning for submitting a reckless graduation application.

Post Conditions	The student has either graduated and exited the system with a degree, or has been warned and remains an active student.
Functional Constraints	A minimum of eight completed classes is required before a graduation application may be submitted. All program-required courses must appear as passed on the student's transcript. A failed graduation application results in one warning issued to the student.

UC-15: Submit a Complaint

Actors	Student, Instructor, Registrar
Trigger	A Student or Instructor submits a formal complaint through the system.
Precondition	The complainant is authenticated and has a grievance against another registered user of the system.
Main Flow	A Student or Instructor submits a formal complaint through the system. The complainant submits a complaint with a written description of the issue, directed to the Registrar. The Registrar reviews the complaint, investigates the situation, and determines the appropriate action. For student complaints against another student or an instructor, the Registrar may issue a warning to the accused party. For instructor complaints against a student, the Registrar must take a concrete action, either punishing the student with a warning or de-registration, or finding the complaint unwarranted and issuing one warning to the complaining instructor instead.
Alternative Flow	If a student accumulates three or more total warnings from any combination of causes, the system automatically suspends them for one semester and requires them to pay a fine to the Registrar.
Post Conditions	The complaint has been resolved with a documented outcome and all applicable warnings or de-registrations have been recorded.
Functional Constraints	Students may file complaints against other students or against instructors. Instructors may file complaints against students only. The Registrar must respond to every instructor complaint with a concrete action and cannot leave it unresolved. A student who accumulates three or more warnings is suspended for one semester and fined.

UC-16: View Instructor Dashboard

Actors	Instructor
Trigger	The Instructor logs in or navigates to their personal dashboard.
Precondition	The Instructor is authenticated and has at least one currently assigned course for the active semester.
Main Flow	The Instructor's dashboard displays all courses currently assigned to them for the active semester. The Instructor can select any course to view its full student roster. For each student on the roster, the system displays the student's basic personal information and complete academic record including grades, GPA, and standing.
Alternative Flow	None.
Post Conditions	The Instructor has reviewed the relevant student information within the scope of their assigned courses.
Functional Constraints	Instructors are strictly limited to viewing academic records of students who are

	currently enrolled in one of their active courses. They may not view records of students in other courses or from previous semesters.

UC-17: View Student Dashboard

Actors	Student
Trigger	The Student logs in or navigates to their personal dashboard.
Precondition	The Student is authenticated and has completed the mandatory first-login password change if applicable.
Main Flow	The student's dashboard displays their currently enrolled courses, all historical grades, current semester GPA, cumulative GPA, any active warnings, and their honor roll status if applicable. The student can navigate from the dashboard to other available features such as course registration or review submission.
Alternative Flow	If it is the student's first login, the system presents a mandatory onboarding tutorial walking them through the key features of the system before the dashboard is displayed.
Post Conditions	The student has viewed their complete personal academic record and system standing.
Functional Constraints	Students may only view their own records and are not permitted to access any other student's information. The onboarding tutorial is mandatory on the first login and cannot be skipped.

UC-18: Registrar Full Administration

Actors	Registrar
Trigger	The Registrar logs in or initiates any administrative action within the system.
Precondition	The Registrar is authenticated with superuser credentials.
Main Flow	The Registrar has unrestricted access to every record, configuration, and function in the system. This includes all student and instructor records, all course data and enrollment information, all review content including reviewer identities, complaint logs, and all system configuration settings. The Registrar manages the transition between the four semester periods, processes all applications, resolves all complaints, reviews justifications, approves or denies graduation requests, and issues all disciplinary actions including warnings, fines, suspensions, and terminations.
Alternative Flow	None.
Post Conditions	The administrative action has been completed and all changes are recorded in the system.
Functional Constraints	The Registrar is the sole superuser of the system and inherits all capabilities of every other role. All semester period transitions are controlled exclusively by the Registrar. Every administrative action that modifies a student or instructor record must be logged for audit purposes.

UC-19: AI-Assisted Q&A

Actors	Visitor, Student, Instructor, Registrar
Trigger	A user types a question into the AI-enabled text area available on their page.

Precondition	The user is on any page of the system that provides the AI text area, and the local vector database and external LLM service are both operational.
Main Flow	The user submits a natural language question. The system first queries the local vector database, which contains College0-specific information not available in general LLMs. If a sufficiently confident answer is found in the local database, the system returns that answer to the user. If no adequate answer is found locally, the system forwards the query to the external LLM, retrieves its response, and returns it to the user along with a clearly visible hallucination warning.
Alternative Flow	None.
Post Conditions	The user has received an answer sourced either from the local vector database or from the external LLM with an appropriate caveat displayed.
Functional Constraints	The local vector database must always be queried before falling back to the external LLM. All LLM-generated responses must be accompanied by a visible hallucination warning. Visitors may only ask general questions about classes and program requirements. Students may additionally ask questions about their currently enrolled courses. Instructors may additionally ask questions about students enrolled in their current courses. Registrars have unrestricted query access with no scope limitations.

3.2 Supplementary Requirements

This section defines the non-functional requirements of the College0 system. These requirements ensure the system operates efficiently, securely, and reliably.

3.2.1 Performance

- The system shall support a minimum of 1,000 concurrent users.
- The average page load time shall not exceed 2 seconds.
- AI query responses shall be returned within 3 seconds.
- Database operations (search, retrieval) shall complete within 1 second.

3.2.2 Security

- All communication shall use HTTPS encryption.
- Passwords shall be stored using secure hashing algorithms (e.g., bcrypt).
- The system shall implement role-based access control (RBAC):
    - Students: access only their own records
    - Instructors: access only assigned students
    - Registrar: full system access
- The system shall protect against:
    - SQL Injection
    - Cross-Site Scripting (XSS)
    - Unauthorized access

3.2.3 Usability

- The interface shall be intuitive and user-friendly.
- Navigation shall remain consistent across all dashboards.
- First-time users shall complete a mandatory onboarding tutorial.
- The system shall be accessible on desktop and mobile devices.

3.2.4 Reliability

- The system shall maintain 99% uptime availability.
- Automatic data backups shall occur daily.
- System recovery time after failure shall not exceed 5 minutes.

3.2.5 Scalability

- The system shall support future growth in users and data.
- New features and modules shall be added without affecting existing functionality.

3.2.6 Maintainability

- The system shall use a modular architecture.
- Code shall be properly documented.
- Updates and maintenance shall not disrupt active users.

3.2.7 AI Requirements

- The system shall query the local vector database before external AI services.
- If external AI is used, responses must include a hallucination warning.
- AI responses shall be clearly labeled and explainable.

4. Supporting Information

This section provides additional resources and references that support the understanding of the system.

4.1 Index

Terms	Meaning
GPA	Grade Point Average
UC	Use Case
AI	Artificial Intelligence
RBAC	Role-Based Access Control

4.2 Appendices

Appendix A: System Architecture

The system consists of:

- Frontend (User Interface dashboards)
- Backend (Application server)
- Database (Student, Course, Review data)
- AI Module (Local vector database + External LLM)

Appendix B: User Interface Prototypes

The system includes the following interfaces:

- Login Page
- Student Dashboard
- Instructor Dashboard
- Registrar Dashboard
- AI Chat Interface

Appendix C: Use Case Diagrams

Actors included:

- Visitor
- Student
- Instructor
- Registrar

These diagrams illustrate system interactions and workflows.

Appendix D: Assumptions

- Users have internet access.
- Students follow academic policies.
- Registrars enforce system rules correctly.

Appendix E: Constraints

- The system depends on:
    - Database availability
    - External AI service reliability
- The system must comply with:
    - Data privacy standards
    - Institutional academic policies